Table 3. Quality, serving cost and capacity constraints. Quality and serving cost of three attention variants at 16k context; the 32k column is for reference and is not ranked.

	Matched 16k context			Reference
Metric	**Dense** 16k	**Sparse** 16k	**Hybrid** 16k	**Hybrid** 32k
Quality (1,000 requests)				
Task success (%) ↑	74.2	74.3	**76.6**	78.0
Calibration error (pp) ↓	4.8	4.2	**3.5**	3.2
Serving (batch 4)				
P95 request latency (ms) ↓	184.2	157.9	**149.8**	230.3
Throughput (requests/s) ↑	30.4	34.7	**36.2**	23.5
Peak device memory (GB) ↓	36.4	30.1	**28.5**	44.8
Energy per request (J) ↓	9.8	8.2	**7.6**	12.4
Capacity stress (batch 32)				
P95 request latency (ms) ↓	*OOM*	392	**351**	*OOM*
Throughput (requests/s) ↑	*OOM*	61.3	**66.9**	*OOM*

Synthetic data. One 80 GB GPU, BF16 weights, 256 output tokens per request. P95 is a within-run latency quantile, not an interval. Bold: best 16k value in each row, following its arrow. OOM = out of memory (an outcome, not a score of zero).